SE 1105 - PROGRAMMING AND PROBLEM SOLVING I

Conditional Statements

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2. Compound Expressions
   - Operator Precedence
3. Compound Statements
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Review

printf in C

```c
#include <stdio.h>

void main(){
   int bYear = 2006;
   printf("I was born in %d.\n",bYear);
   printf("I am %d years old.\n",2024-bYear);
   printf("%d is divisible by %d.\n", bYear,2);
}
```

Review

Function Definition and Calling

```
#include <stdio.h>

int square(int a)
{
    return a*a;
}

void main(){
    int n = 4;
    int p = square(n);
    printf("Square of %d is %d\n",n,p);
}
```

Review

Extract a 4-Digit Number

```
1  void printDigits(int number)
2  {
3    int digit1 = number / 1000;
4    number = number - (digit1 * 1000);
5    int digit2 = number / 100;
6    number = number - (digit2 * 100);
7    int digit3 = number / 10;
8    int digit4 = number % 10;
9    printf("%d\n%d\n%d\n%d\n",digit1,digit2,
   digit3,digit4);
10 }
```

Review

Extract a 4-Digit Number [Version 2]

```
1  void printDigits_v2(int number)
2  {
3    int digit4 = number % 10;
4    number = number / 10;
5    int digit3 = number % 10;
6    number = number / 10;
7    int digit2 = number % 10;
8    int digit1 = number / 10;
9    printf("%d\n%d\n%d\n%d\n",digit1,digit2,
   digit3,digit4);
10 }
```

Review

Boolean Operators

```
1  void logicalOperators( )
2  {
3     bool p=false , q=false;
4     printf("%d and %d = %d\n",p,q,p && q);
5     p = false; q = true;
6     printf("%d and %d = %d\n",p,q,p && q);
7     p = true; q = false;
8     printf("%d and %d = %d\n",p,q,p && q);
9     p = true; q = true;
10    printf("%d and %d = %d\n",p,q,p && q);
11 }
```

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Compound Expressions

- When an expression contains more than one operand (other than the assignment operand) then it is called a **compound expression**.
- e.g. f = a - b++ * c + d % 2;
- The precedence of the operands determines the order of applying them in compound expressions.

Operator Precedence

Precedence	Operator
1	( ) and prefix ++ , −−
2	* , / , %
3	+ , −
4	Postfix ++ , −− *(Note: Applied right after arithmetic operators in 2 and 3)*
5	< , <= , > , >=
6	== , ! =
7	&&
8	||
9	= , and other assignment operators

Back to the previous example: f = a - b++ * c + d % 2;

Exercise

- Write a function that receives the lab, midterm, and final scores of a student and returns his or her grade according to the assessment of our course.
- Write a function that converts a Fahrenheit degree value to Celcius.

$$^{\circ}C = (^{\circ}F - 32) * \frac{9}{5}$$

- Write a function that returns the value of the following logical statement.

$$(p \vee q) \wedge \sim (p \wedge (q \vee r))$$

NOTE: Since we only covered **bool** and **int** data types, assume that the numerical calculations will be done with integers.

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Compound Statements

- Block of statements surrounded by { ... }.
- Controls the scope of the variables.
- Generally used for functions, conditional statements, and loops.

```
1 #include<stdio.h>
2 void main( )
3 {
4   int a = 10;
5   printf("outer a: %d\n",a);
6   {
7       int a = 5;
8       printf("inner a: %d\n",a);
9   }
10  printf("outer a: %d\n",a);
11 }
```

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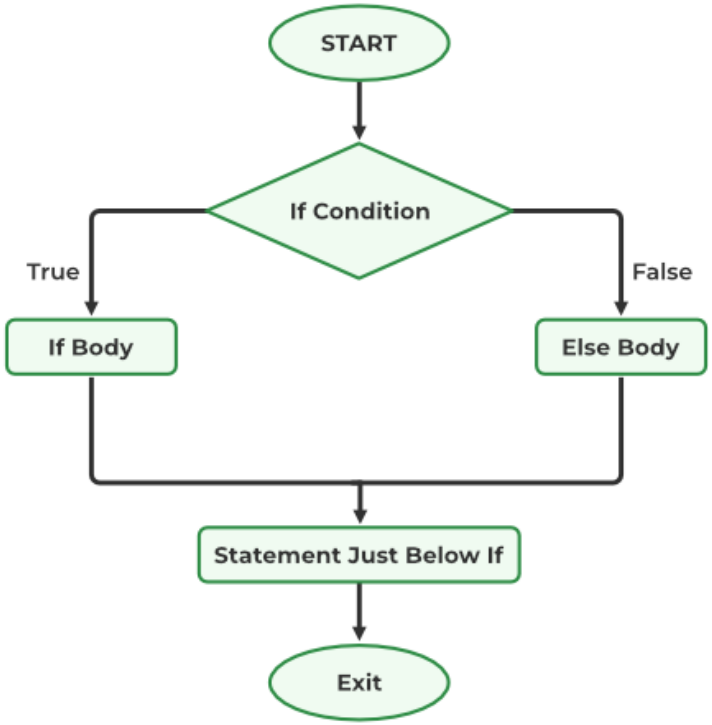
Conditional Statements

- Decision-making is a crucial part of all programs.
- For these cases, **if** conditional statement in C is used.
- A simple **if** statement in C contains **if** keyword, condition, and body of code.




```
if ( conditions )
{
    ...statements...
}
else
{
    ...statements...
}
```

Figure:
https://www.geeksforgeeks.org/decision-making-c-cpp/

Conditional Statements

Passing Check

```
1  void checkPassing(int lab, int mt, int fnl)
2  {
3     int grade = (lab + mt*4 + fnl*5) / 10;
4     if ( grade >= 50 )
5     {
6        printf("Passed\n");
7     }
8     else
9     {
10       printf("Failed\n");
11    }
12 }
```

Exercises

1. Write a function that determines the parity of the given integer.
2. Write a function that takes an integer number and returns its absolute value.
3. Write a function that takes three integers and returns the maximum of them.
4. Write a function that receives three integers and checks whether they form a right triangle in Euclidean geometry.

Sequential Conditionals and Nested Conditionals

- Sometimes, the decision-making needs to be done among more than two options.
- In those cases, we either
  - use sequential **if-else if** structure, or
  - use nested **if** structure.

Sequential Conditionals and Nested Conditionals

```
if ( conditions_1 )
{
   ...statements...
}
else if ( conditions_2 )
{
   ...statements...
}
...
else
{
   ...statements...
}
```

```
if ( conditions_1 )
{
   if ( conditions_1_1 )
   {
      ...statements...
   }
}
else if ( conditions_2 )
{
   if ( conditions_2_1 )
   {
      ...statements...
   }
}
...
else
{
   ...statements...
}
```

Sequential Conditionals and Nested Conditionals

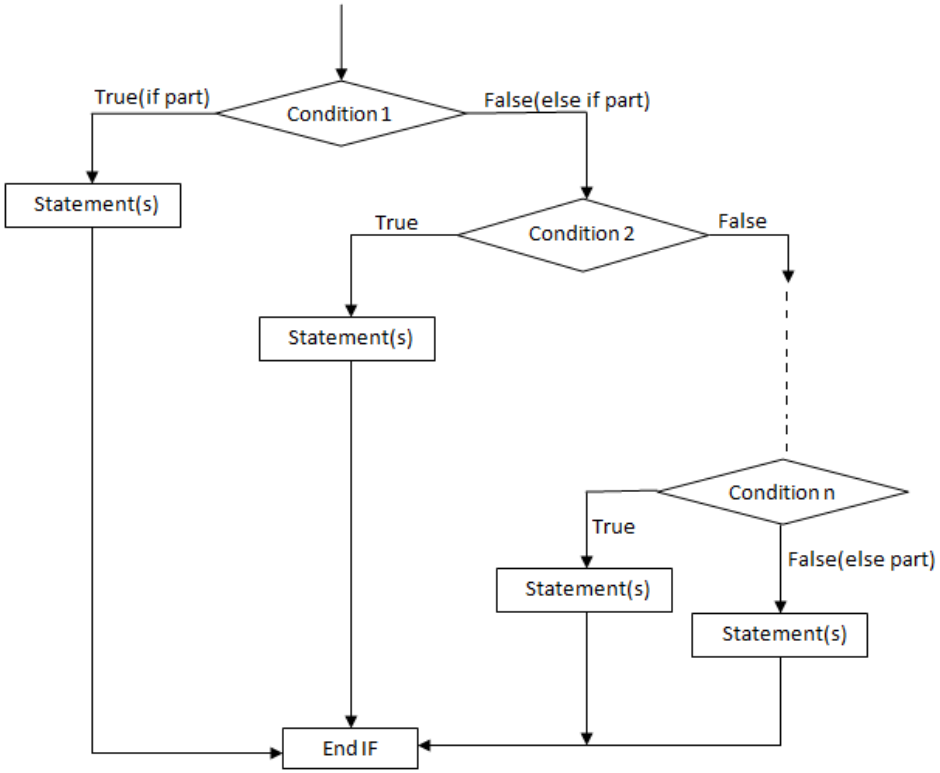


Figure: https://www.programtopia.net/c-programming/docs/if-statements

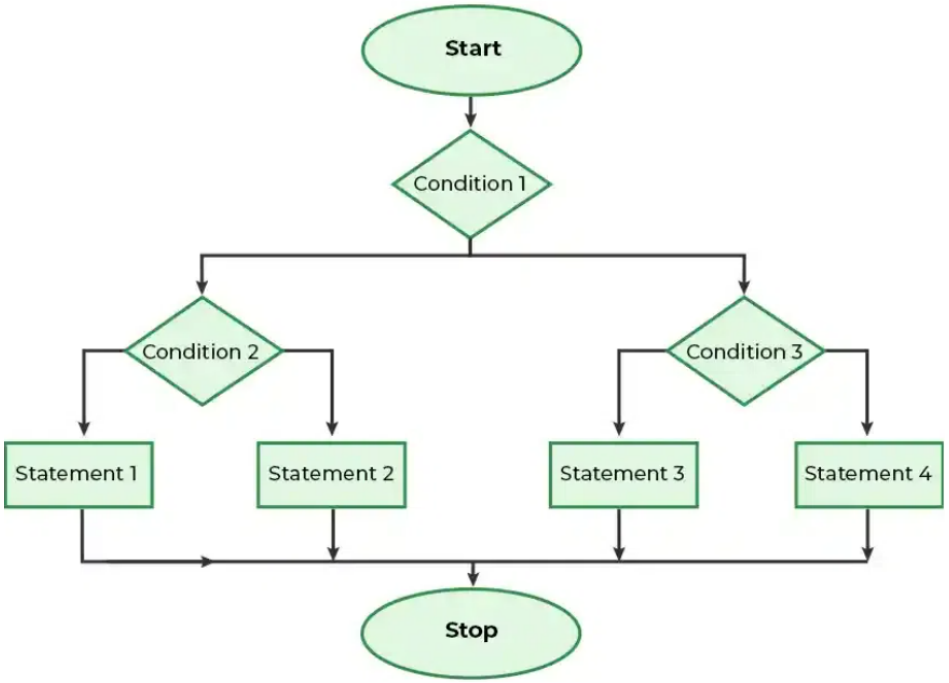


Figure: https://www.geeksforgeeks.org/cpp-nested-if-else-statement/

Sequential Conditionals and Nested Conditionals

Sign of a Number

```
1  void sign(int number){
2    if ( number > 0 )
3    {
4      printf("%d is positive.\n",number);
5    }
6    else if ( number < 0 )
7    {
8      printf("%d is negative.\n",number);
9    }
10   else
11   {
12     printf("%d is neither positive nor
   negative.\n",number);
13   }
14 }
```

Exercises

1. Write a function that finds the maximum of three integers in a nested if structure.
2. Write a function that calculates the grade of a student for this class and prints out his or her letter grade.
3. Write a function that calculates the following piecewise function and returns its result.

$$f(x) = \begin{cases} -x^2 + 3x, & x \leq -4 \\ 5x - 7, & -4 < x \leq 21 \\ \frac{x^3-1}{x+1}, & x > 21 \end{cases}$$